

Project Title: Exploratory data analysis

Objective: to gain a deeper understanding of the data's structure, identify patterns, and uncover insights that can inform subsequent data analysis and modeling tasks, ultimately aimed at predicting passenger survival.

Tools & sources:

Tools: python libraries such as pandas, matplotlib, seaborn for EDA.

Sources: Used titanic dataset.csv file from kaggle

Methodology/ steps:

1. Import python libraries such as:

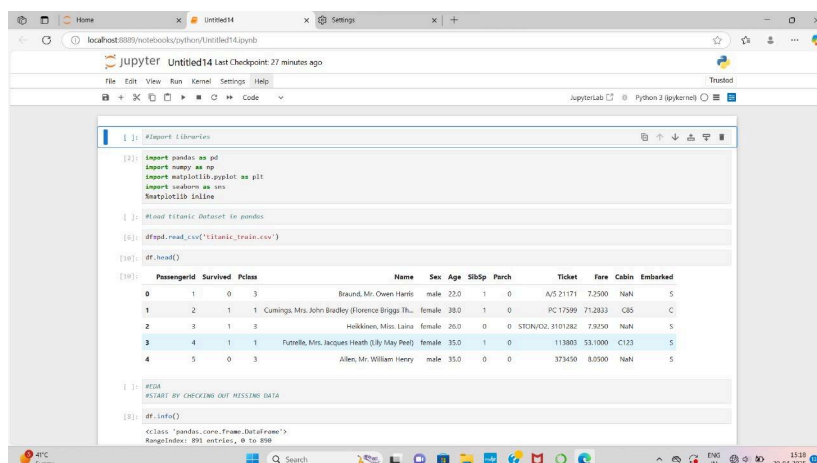
- pandas: used for reading csv file, data cleaning, manipulation analysis.

- seaborn: seaborn is a library for creating statistical graphics in Python. like charts, plot graphs.

- Matplotlib: used for creating interactive and animated visualization.

- %matplotlib inline: allows us to interact with your plots directly in the notebook or shell.

2. Load the dataset in pandas libraries to Read CSV file, as pd.read_csv().



```
[1]: #Import Libraries
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

[3]: #Load Titanic Dataset in pandas
[4]: df=pd.read_csv('titanic_train.csv')
[5]: df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2832	C85	C
2	3	1	3	Halkinnen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

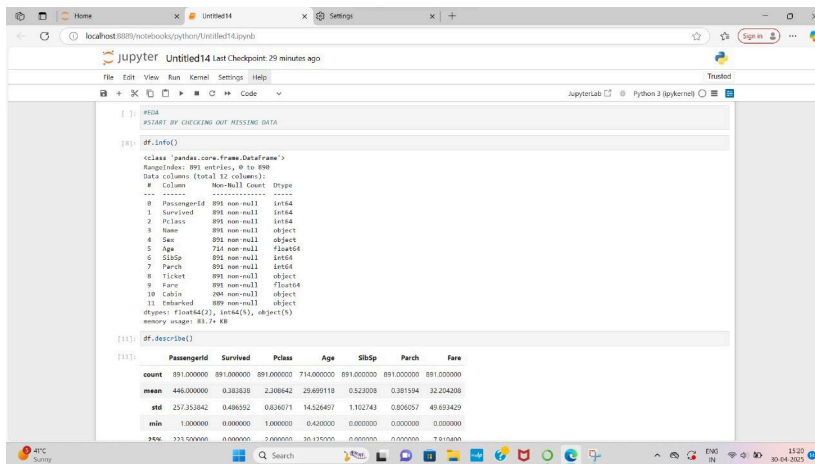
```
[6]: #EDA
#START BY CHECKING OUT MISSING DATA
[7]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
```

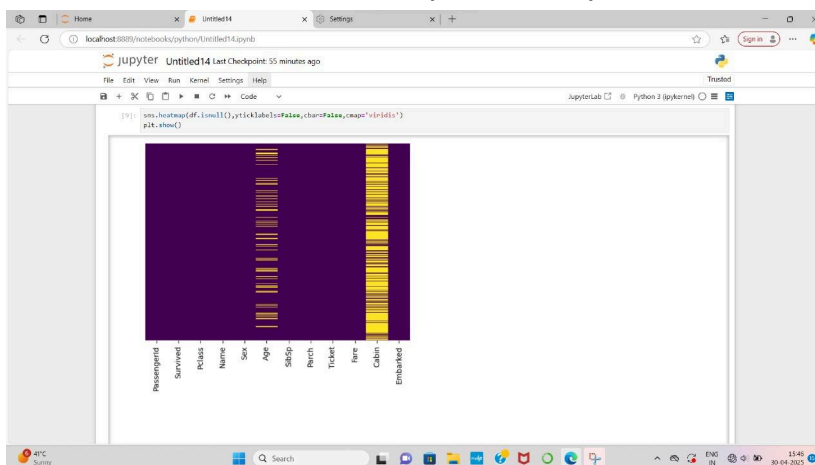
3. Exploratory data analysis:

Start by checking missing data such as

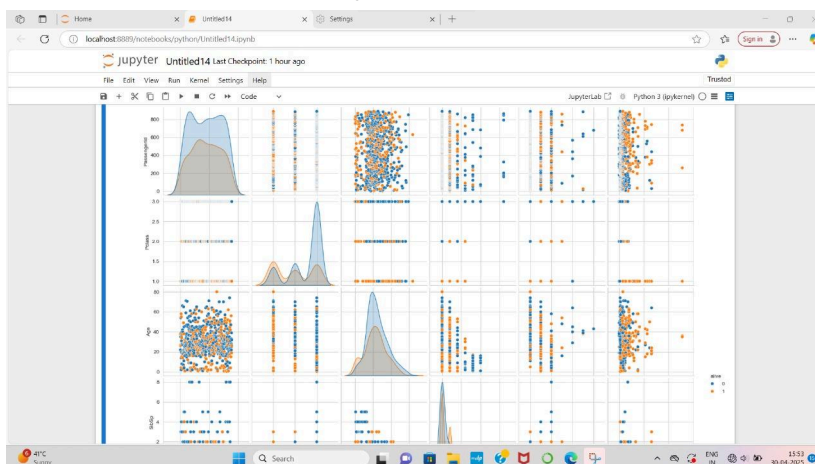
df.info(), df.isnull(), df.describe(),
df.count_values().



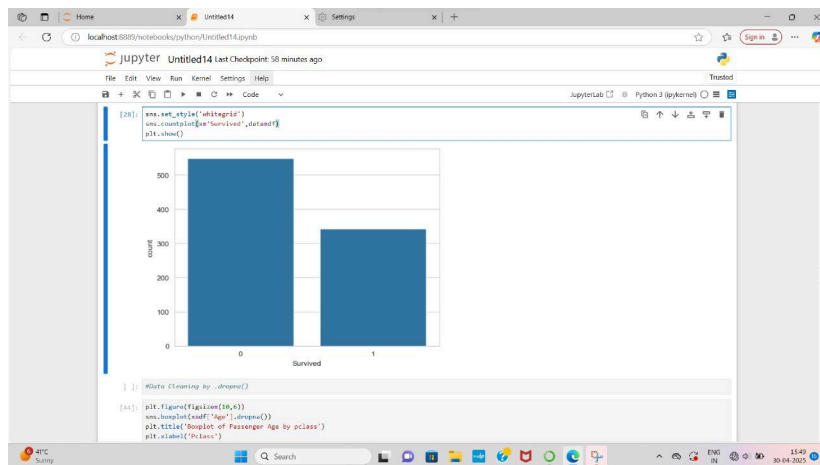
4. Used seaborn to create simple heatmap to see where are missing data



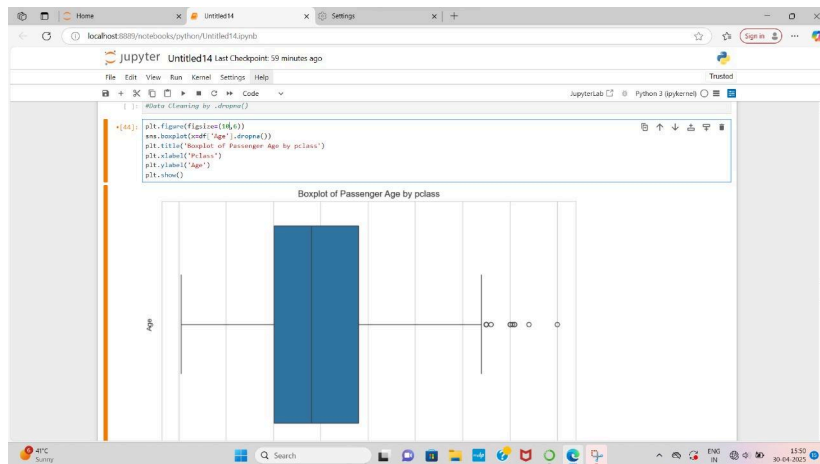
5. created a pairplot by alive columns to colour points by survival



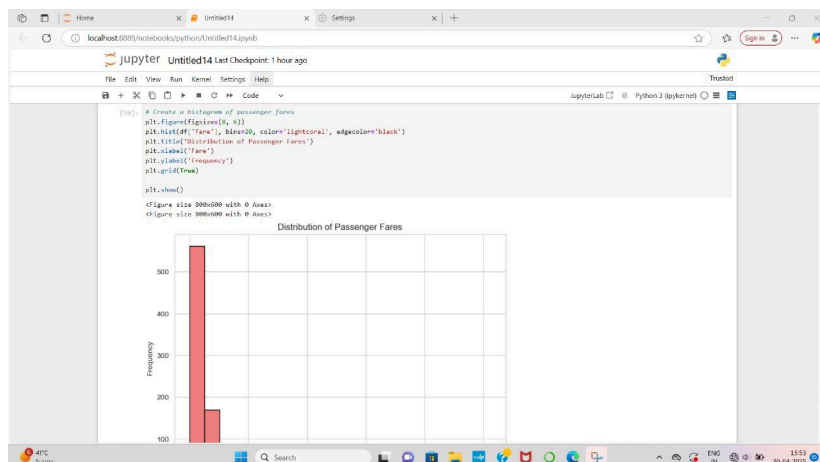
6. Created a countplot based on survived column where 0 indicate not survive while 1 indicate person is survive

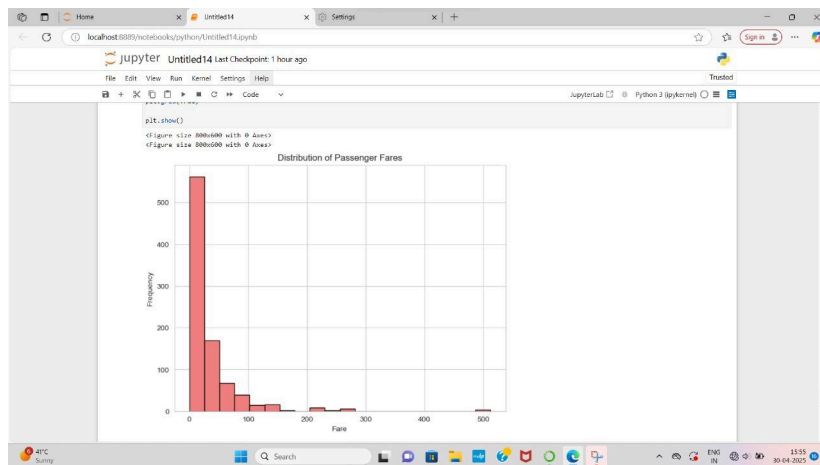


7. created boxplot of passenger age by pclass and data cleaning by drop.na()

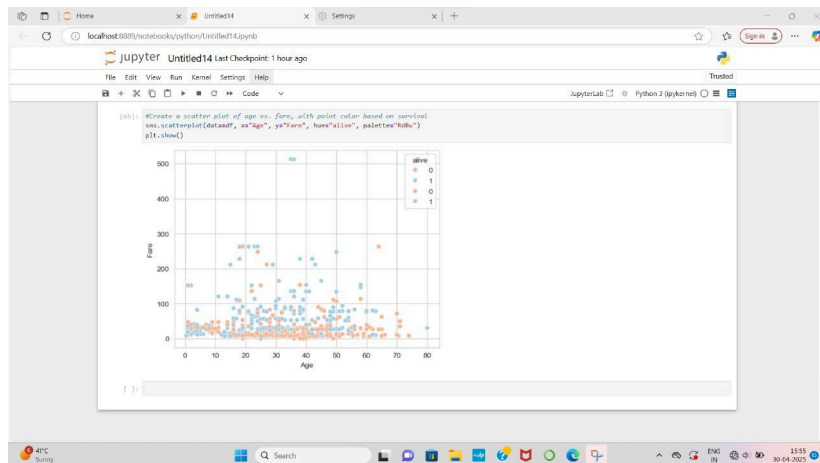


8. created histogram of passenger fares





9.created a scatterplot of age vs fare with point color based on survival



Conclusion & insights:

Analysis of the Titanic dataset reveals key factors influencing passenger survival.

Women had a significantly higher survival rate than men.

Passengers in higher classes (1st and 2nd) were more likely to survive than those in 3rd class.

Children also had a higher survival rate, while elderly passengers had lower chances.

The number of siblings/spouses and parents/children aboard did not show a strong correlation with survival.

This suggests that gender, passenger class, and age were the primary determinants of survival on the Titanic.